

**MATERIAL TEST REPORT**

FFI-QACON-F0003A

Revision: 00, 08/21/2020

Mill Test Report			
DNOV L.P.		Purchase Order:	66268198
PO BOX 40985		Sales Order:	341638
HOUSTON TX 77240		Line Item:	2
		Qty:	18

Item Code:	FF 413AG J0 4A	Material:	A694 F52 LATEST EDITION
Description:	10 900# RF WN XHY FLANGE ANSI B16.5		Manufactured in the U.S.A.
FID:	H12600	Heat Code:	2402112

Element		Ladle	Product
C	Carbon	0.1800	0.1800
Mn	Manganese	1.2600	1.2400
P	Phosphorous	0.0100	0.0090
S	Sulphur	0.0020	0.0010
Si	Silicon	0.2300	0.2300
Cu	Copper	0.0000	0.0100
Ni	Nickel	0.0000	0.0200
Cr	Chromium	0.0000	0.0300

Element		Ladle	Product
N	Nitrogen		
Mo	Molybdenum	0.0000	0.0100
V	Vanadium	0.0000	0.0180
CbNb	Columbium/Niobium	0.0000	0.0130
Ti	Titanium		
Al	Aluminum		
B	Boron	0.0000	0.0001
Sn	Tin		

Additional Chemicals	

Melting Practice	EFVR	Reduction Ratio	>3
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Mechanical Test Results			
Tensile (PSI)	84,121		
Yield (PSI)	61,205		
Elong (%)	25.00		
Red (%)	750.00		
CE	0.3900		
PCM			
HBW	167	to	169
CVN Test Temperature	-50		°F
Ft/Lbs.	169	175	238
Shear (%)	75	75	85
MLE	1.70	1.80	2.40
Charpy Sample	10 X 10 X 55mm		
QTC Type & Size			

Heat Treatment Information	
Austenitizing Temperature (°F)	1670
Austenitizing Time (HRS)	2.5
Cooling Medium (TYPE)	Water
Starting Temperature (°F)	57
Ending Temperature (°F)	104
Tempering Temperature (°F)	1022
Tempering Time (HRS)	3

Additional Comments	
MADE TO FINE GRAIN PRACTICE	


ROLANDO PASTRAN - Quality Representative

4/15/2025
Date

Product is compliant with NACE MR0175/ISO 15156. / Product is compliant with NACE MR0103.
No weld repair is performed.
The product is free of Mercury contamination.
Tensile test is taken from round specimen.
Yield Strength is determined using 0.2% offset method, unless otherwise specified.
When reported on the MTR: Tensile testing per ASTM E 8, Charpy impact testing per ASTM E 23 and Brinell Hardness Testing per ASTM E 10.
Federal Flange manufactures products according to a certified QMS conforming to ISO 9001:2015.
Products are manufactured in compliance to EN 10204 Type 3.1.

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